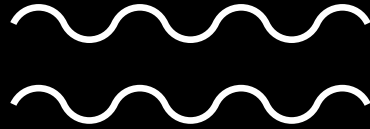


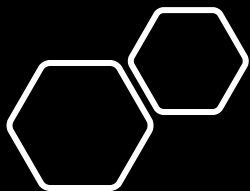
Data Structures in R

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Recap

- What is R? And why learn R?
- Familiarity with the RStudio interface (opened a file)
- Variable assignment and computations in R
- Basic data types in R



What we will cover today

Data Structures in R

Missing values in R

Functions and Data Objects

- A function is a chunk of codes that conducts a specific task. R comes with many built in functions (e.g. mean, summary, t.test, cor.test, and so on)

Use help to look up these functions:

<-

read.csv

write.csv

plot

print

Functions and Data Objects

- Data comes in many types recognized by R and RStudio
- The most common types are:

Data Type	Also Known as	Example
Numeric	Float	10, 2.1, -0.9
Character	String or text	“Calgary”, “Ottawa”, “Intervention”, “Control”
Logical	Boolean	TRUE, FALSE

Data Structures in R

Data Structures in R: Vectors

- A **vector** is a sequence/collection of data elements of the same basic type
- Members in a vector are officially called **components**.
- To assign multiple values to a variable, we can use an R object called a '**vector**'
- To create a vector, you should use **c()** function which means to combine the elements into a vector. **For Example:** *vect = c(14,26,38,30)*
- To access a specific element in the vector, we simply need to call *variable_name[i]*, '*i*' being the element's position in the vector. **For example:** *vect[3]* would return *38*

Data Structures in R: Matrices

- A matrix is a sequence/collection of data elements of the same basic type arranged in a two-dimensional rectangular layout.
- A matrix is created using the **matrix()** function

```
my_matrix <- matrix(svect1, nrow=5, ncol=2)
my_matrix
```
- Being a 2-dimensional object, in order to obtain a specific value within the matrix, 2 coordinates needs to be entered.

Data Structures in R: Matrices

`my_matrix[i,j]` would return the element on the *ith* row, in the *jth* column

`my_matrix[i, ]` would return the entire *ith* row

`my_matrix[, j]` would return the entire *jth* column

Extract rows and columns (vectors), and cells from **matrices** using square brackets [row, col].

- For example, if you would like to extract all the values of the 3rd row in

my_matrix, you can use `my_matrix[3,]`.

Extract from `my_matrix` 1) the 2nd column, 2) the 5th row, and 3) cell that is in the 3rd row and 1st column

`my_matrix[,2]; my_matrix[5,]; my_matrix[3,1]`

Data Structures in R: Dataframes

- A data frame is used for storing data tables.
- It is a list of vectors of equal length. Unlike matrices, it can gather vectors containing **different** basic types
- Selection of specific elements in data frames works the same way as for matrices.
 - **For example:** `my_dataframe[i,j]` would return the element on the *ith* row, in the *jth* column

Lists

- A list in R allows you to gather a variety of objects under one name (that is, the name of the list) in an ordered way.
- These objects can be matrices, vectors, data frames, even other lists, etc. It is not even required that these objects are related to each other.
- To create a list on R, use the `list()` function.
- To access the *ith* object in the list, write `my_list[[i]]`
- If you want to access a variable in the *ith* object in the list, write `my_list[[i]][variable coordinates]`.

Missing Values in R

Missing values are denoted by NA or NaN for undefined mathematical operations.

- `is.na()` is used to test objects if they are NA
- `is.nan()` is used to test for NaN
- A NaN value is also NA but the converse is not true

Other Resources

- <https://swcarpentry.github.io/r-novice-inflammation/13-supply-data-structures/>
- <https://www.geeksforgeeks.org/data-structures-in-r-programming/>
- <https://data-flair.training/blogs/data-structures-in-r/>
- https://warwick.ac.uk/fac/sci/moac/degrees/moac/ch923/r_introduction/help/
- <https://towardsdatascience.com/dealing-with-missing-data-in-data-science-projects-e8ac7a4efdff>
- <https://jameshoward.us/2016/07/18/nan-versus-na-r/>
- <https://www.r-bloggers.com/2012/08/difference-between-na-and-nan-in-r/>
- **Important:** <https://cran.r-project.org/doc/manuals/r-release/R-lang.pdf>

**to be
continued...**

thanks for listening.

Thank You

